

# Cloud Volume Management System

In this project, I have created a instance with certain volume and added extra volume made it visible on both Linux and windows instances.

=====

How to create a volume and Allocate to the instance

While creating a new instance in configure storage tab click on "Add new volume"

▼ **Configure storage** [Info](#) [Advanced](#)

1x  GiB  ▼ Root volume (Not encrypted)

ⓘ

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

×

Add new volume

The selected AMI contains more instance store volumes than the instance allows. Only the first 0 instance store volumes from the AMI will be accessible from the instance

Add extra volume and launch instance

▼ Configure storage [Info](#)

Advanced

1x

30

GiB

gp2

▼

Root volume (Not encrypted)

1x

4

GiB

gp3

▼

EBS volume (Not encrypted)

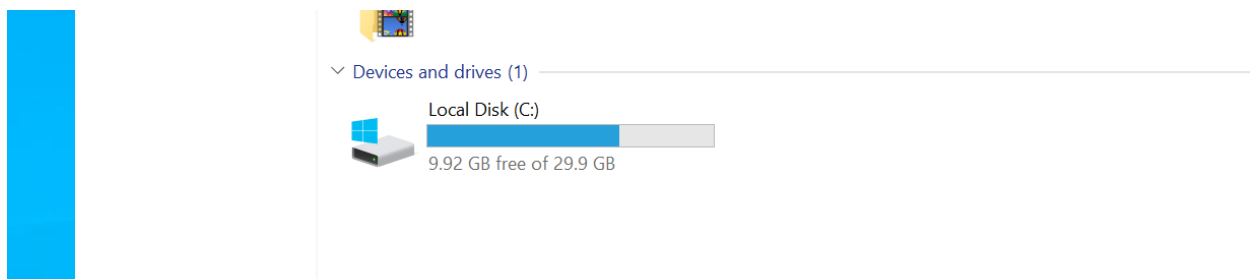
Remove

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

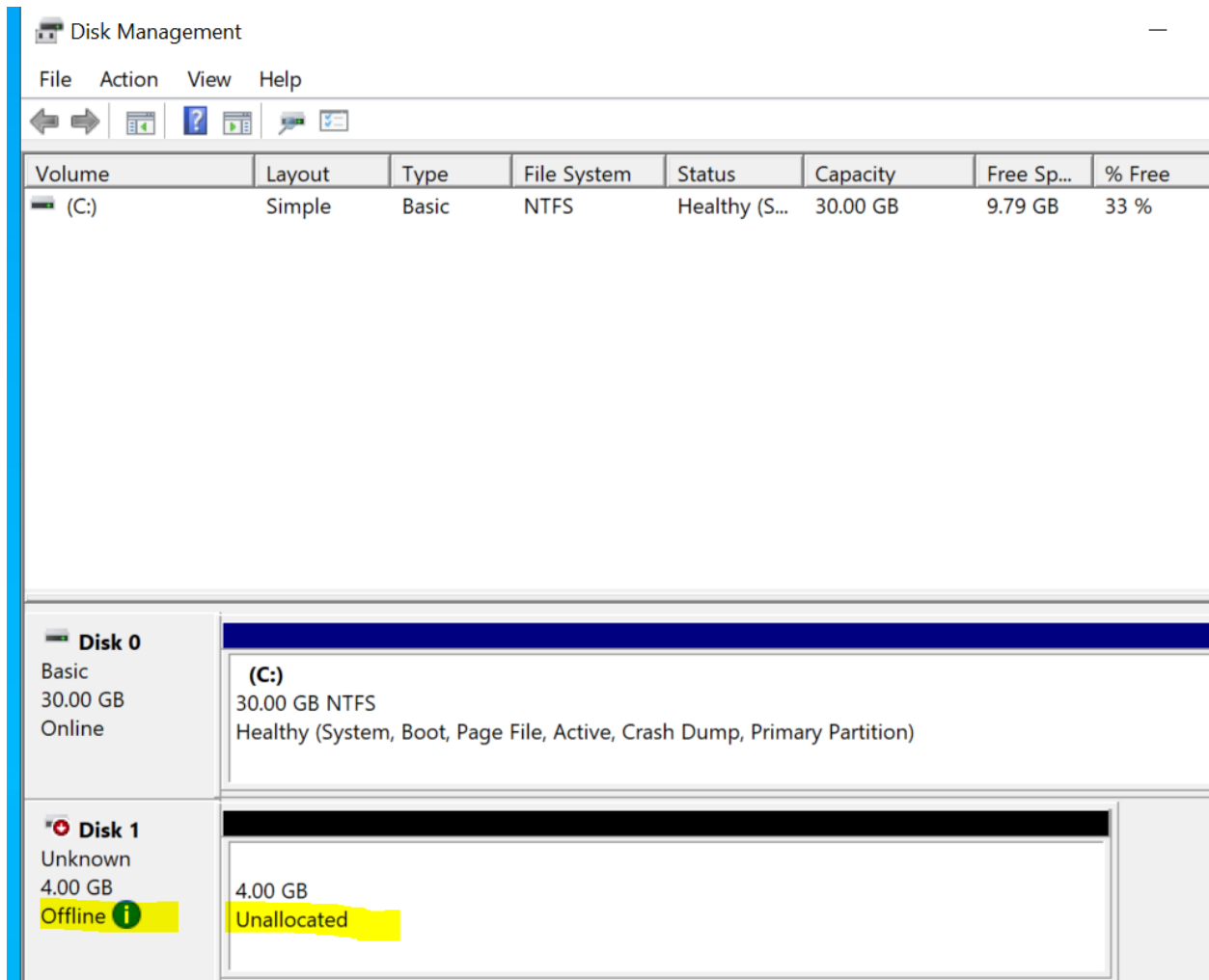
×

Now Connect the instance to allocate the storage

Here you can see the storage of 30 gb only because, the 4gb we added is unallocated



To allocate the 4gb, Right click on 'Start button' -> Disk management. Now you can see the storage is unallocated and offline



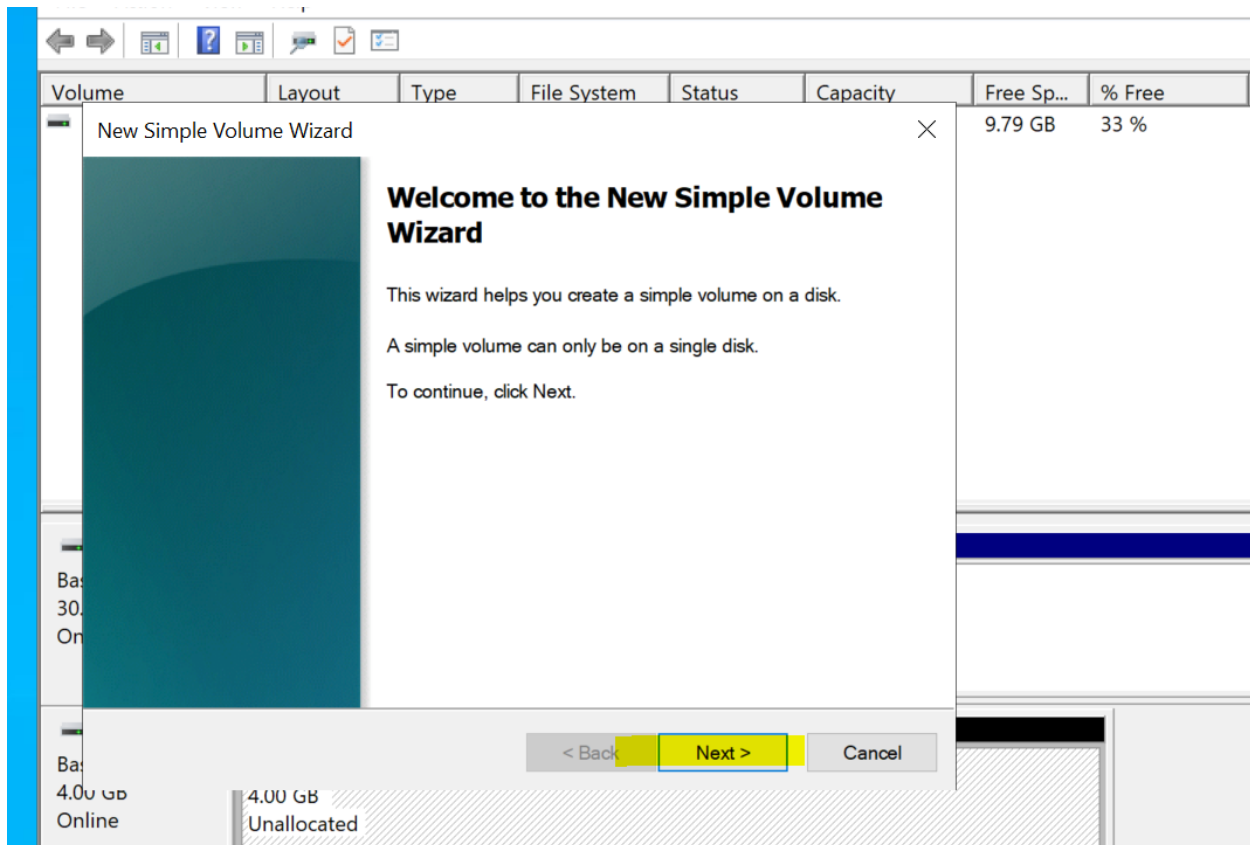
First click on offline and make it online

After making it online, now it will show not initialized

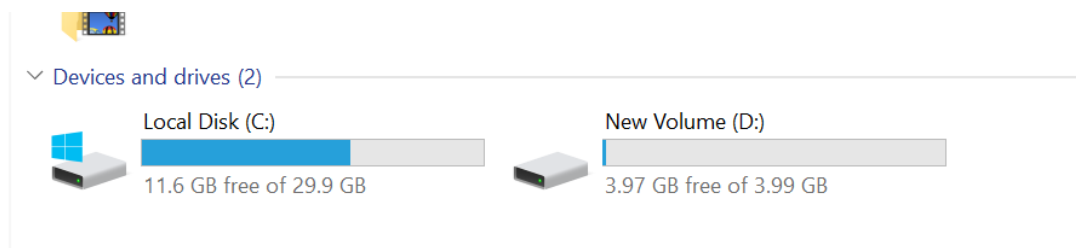
To initialize it click on not initialized and make it initialized



Now Right click on the unallocated 4gb -> new simple storage

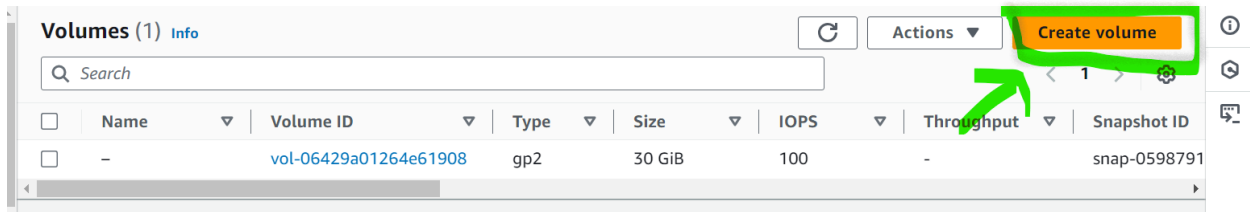


Click on next and finish the process, Now the storage will be allocated and it will show in disk.



Create a new volume and attach it to the existing instance

Click on the create volume button



Select the type and size of the volume

## Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

### Volume settings

Volume type [Info](#)

General Purpose SSD (gp3) ▼

Size (GiB) [Info](#)

100

Min: 1 GiB, Max: 16384 GiB. The value must be an integer.

IOPS [Info](#)

Now give the details based on the requirement

|                   |                                                                                       |
|-------------------|---------------------------------------------------------------------------------------|
| IOPS              | Input output operation that can be supported by the volume                            |
| Throughput        | Performance of the volume                                                             |
| Availability zone | It has to be same with the availability zone of the instance you are going to connect |
| Snapshot          | If you want you can create or else you can create it separately also                  |

IOPS [Info](#)

3000

Min: 3000 IOPS, Max: 16000 IOPS. The value must be an integer.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Availability Zone [Info](#)

us-east-1a

Snapshot ID - *optional* [Info](#)

Don't create volume from a snapshot



Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with y

Click on create button

**Snapshot summary** [Info](#)



🕒 Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel

Create volume

We have created a new volume, now we're going to connect to the instance

Click on the volume ID

| <input type="checkbox"/> | Name | Volume ID             | Type | Size   | IOPS | Throughput | Snapshot ID  |
|--------------------------|------|-----------------------|------|--------|------|------------|--------------|
| <input type="checkbox"/> | -    | vol-06429a01264e61908 | gp2  | 30 GiB | 100  | -          | snap-0598791 |
| <input type="checkbox"/> | -    | vol-0fe15230db44d40fb | gp3  | 11 GiB | 3000 | 125        | -            |

Go to Actions => Attach volume

[illegible]

Select the instance. Select the device name and click on attach volume  
You have attached the volume to the instance, go and allocate it as we did above

## How to modify or add extra volume in the existing volume of an instance

Open the volume and click on **modify**

EC2 > Volumes > vol-0fe15230db44d40fb

**vol-0fe15230db44d40fb**
↺
Actions ▼
Delete
Modify

|                                    |                |             |                                    |
|------------------------------------|----------------|-------------|------------------------------------|
| Volume ID<br>vol-0fe15230db44d40fb | Size<br>11 GiB | Type<br>gp3 | Volume status<br>Insufficient data |
|------------------------------------|----------------|-------------|------------------------------------|

You can only increase the previous volume it cannot be decreased  
Change the **size** of the **volume**,  
Click on **modify**

General Purpose SSD (gp3)

Size (GiB)
Info

11

Min: 1 GiB, Max: 16384 GiB. The value must be an integer.

IOPS
Info

3000

Min: 3000 IOPS, Max: 16000 IOPS. The value must be an integer.

Throughput (MiB/s)
Info

125

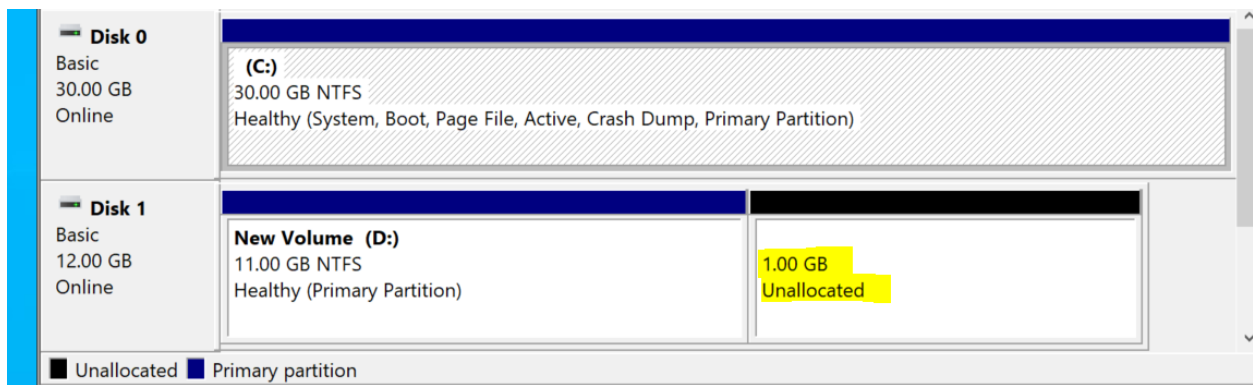
Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Cancel
Modify

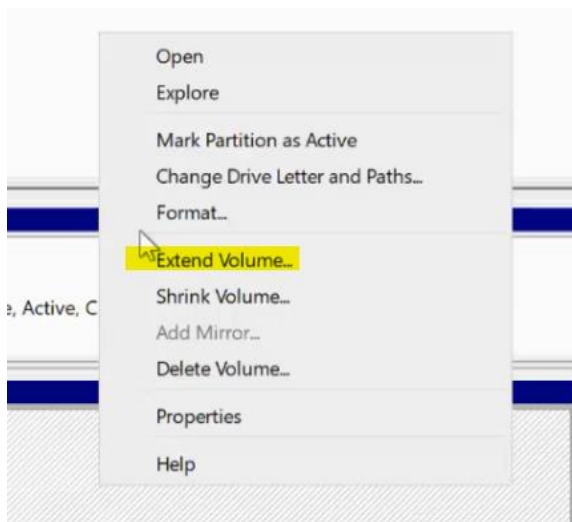
The previous volume will be modified but it will be in unallocated state, you have change that into allocated state as we did previously in the document but we are going to extent the volume with the modified volume  
I have changed the volume size from 11gb to 12 gb

|  | Name ▼ | Volume ID ▼           | Type ▼ | Size ▼ | IOPS ▼ | Throughput ▼ | Snapshot  |
|--|--------|-----------------------|--------|--------|--------|--------------|-----------|
|  | -      | vol-06429a01264e61908 | gp2    | 30 GiB | 100    | -            | snap-0598 |
|  | -      | vol-0fe15230db44d40fb | gp3    | 12 GiB | 3000   | 125          | -         |

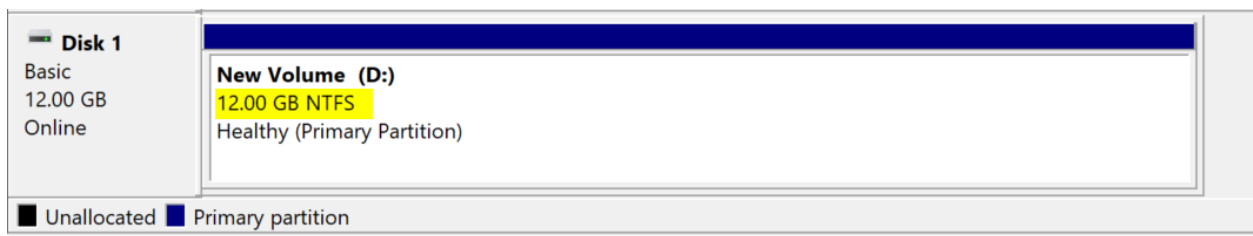
You can see the extra 1gb in unallocated state in the disk management



Right click on the 11gb -> click on extend volume -> Finish



Now you can see that 11gb -> 12gb



=====

**How to connect volume to an linux instance**

Create an linux instance and connect it with putty

Now take a look at the disk sections by using this command

"df -h" [disk free human readable]

After that give this command "**lsblk**" list block to see the unmounted volume in the instance.

```
root@ip-172-31-29-145:~  
[root@ip-172-31-29-145 ~]# lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT  
└─xvda      202:0    0   8G  0 disk  
   └─xvda1  202:1    0   8G  0 part /  
└─xvdb      202:16   0   7G  0 disk /luffy  
xvdc        202:32   0   3G  0 disk  
[root@ip-172-31-29-145 ~]#
```

Now we're going to make a file system and directory to mount the extra volume

These are the commands to make a file system and directory

Mkfs -t ext4 <path of unmounted volume>

Make files system

-t = type of the file

Ext4 = format of the file

Mkdir /<directory name>

After that we're going to mount the volume

Mount <file system path> <directory path>

Mount /dev/xvdc /usoop

```

tmpfs          96M      0   96M    0% /run/user/1000
[root@ip-172-31-29-145 ~]# cd
[root@ip-172-31-29-145 ~]# mount /dev/xvdc /usoop
[root@ip-172-31-29-145 ~]# df -h

```

| Filesystem | Size | Used | Avail | Use% | Mounted on     |
|------------|------|------|-------|------|----------------|
| devtmpfs   | 467M | 0    | 467M  | 0%   | /dev           |
| tmpfs      | 477M | 0    | 477M  | 0%   | /dev/shm       |
| tmpfs      | 477M | 444K | 476M  | 1%   | /run           |
| tmpfs      | 477M | 0    | 477M  | 0%   | /sys/fs/cgroup |
| /dev/xvda1 | 8.0G | 1.9G | 6.2G  | 23%  | /              |
| /dev/xvdb  | 6.8G | 24K  | 6.4G  | 1%   | /luffy         |
| tmpfs      | 96M  | 0    | 96M   | 0%   | /run/user/1000 |
| /dev/xvdc  | 2.9G | 24K  | 2.8G  | 1%   | /usoop         |

```

[root@ip-172-31-29-145 ~]#

```

We can also modify the volume like we did in the windows system

Change the volume size and enter this command

```
"resize2fs /dev/xvdc /usoop"
```